Relative Depth-Guided Cross-Domain Semantic Segmentation for High-Resolution Satellite Remote Sensing Images

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***Abstract*—Satellite remote sensing images exhibit significant cross-domain discrepancies due to variations in sensors, imaging conditions, and regional characteristics. These discrepancies degrade the performance of semantic segmentation models on unseen target domains. Unsupervised domain adaptation (UDA) mitigates this problem but requires target-domain images during training. Achieving robust cross-domain generalization (DG) with foundation models typically relies on large-scale architectures, incurring substantial computational overhead that limits practical deployment. This article introduces a novel perspective by leveraging zero-shot relative depth maps as domain-invariant geometric priors to guide cross-domain semantic segmentation of satellite remote sensing images. The proposed framework is compatible with most encoder–decoder architectures and makes three key contributions: 1) a unified depth-guided framework for optical remote sensing cross-domain segmentation; 2) a multiscale contrastive learning module that aligns optical and relative depth features, enhancing optical feature representation; and 3) a refinement strategy using shallow relative depth features to guide semantic mask prediction. Experiments across three cross-domain scenarios, including the LoveDA benchmark, a self-constructed circumpolar cross-seasonal dataset, and a cross-dataset generalization setting, demonstrate that relative depth serves as an effective spatial prior. Notably, the proposed method achieves comparable or superior accuracy to state-of-the-art UDA approaches that require target-domain images during training, while operating under the stricter zero-shot DG setting where no target data is accessible. The codes will be released at https://github.com/ArthasMil/DGCD.**

***Index Terms*—Cross-domain, high-resolution, relative depth, satellite remote sensing images, semantic segmentation.**

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I. Introduction

SEMANTIC segmentation assigns a class label to each pixel of an image [1]. Compared with scene classification [2] and object detection [3] in remote sensing, this pixel-level task is inherently more challenging. Satellite image semantic segmentation facilitates global surface mapping and supports applications including land cover classification [4], large-scale cartography [5], and urban monitoring [6], making it a prominent research focus in remote sensing [7].

Despite extensive research [8], satellite optical images vary significantly in resolution, color, texture, and viewpoint due to differences in sensor types, imaging conditions, and geographical regions. This cross-domain phenomenon introduces a distribution shift between training data (source domain) and deployment data (target domain), substantially degrading model performance. Two primary directions have been explored to address this challenge: domain adaptation (DA) and domain generalization (DG).

DA transfers knowledge from a source domain to a target domain with a different distribution [9]. Existing DA methods fall into four categories: distribution measurement based [10], [11], [12], adversarial learning based [13], [14], self-training based [15], [16], [17], [18], [19], [20], [21], and large vision model (LVM) based [22], [23]. These approaches reduce the distribution gap between labeled source domains and unlabeled target domains via transfer learning. However, standard DA is limited to prespecified target domains, making it inapplicable to unseen domains.

DG methods require no access to target-domain images, enabling direct inference on unseen domains with greater practical utility. However, classical DG approaches still struggle to generalize robustly across diverse satellite remote sensing cross-domain scenarios [24], [25], [26], [27], [28]. Recently, LVMs with strong generalization ability have been introduced to improve performance [23], [29].

Current DG methods follow an image-to-image generalization paradigm. However, in satellite-based Earth observation, object-space geometric information such as depth and elevation provides a crucial reference for semantic understanding. As illustrated in Fig. 1, building edges serve as both semantic boundaries and depth variation boundaries: semantic transitions correspond to depth shifts, while semantically continuous regions exhibit smooth depth distributions, revealing an

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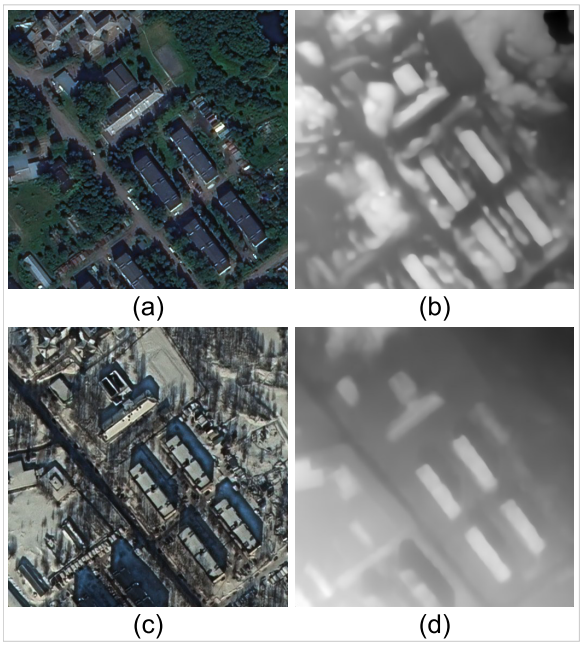


Fig. 1. Dual role of building edges as semantic boundaries and depth variation boundaries, highlighting the inherent consistency between rDepth and semantics. Optical image in (a) summer and (c) winter, rDepth map in (b) summer and (d) winter.

inherent consistency between semantics and depth. Nevertheless, acquiring depth maps from optical images typically requires stereo image collection followed by triangulation and postprocessing, limiting practical applicability.

The depth anything model (DAM) [30], [31], [32] offers a practical solution for depth acquisition. As a vision foundation model, it generates zero-shot relative depth (rDepth) maps for any input image, capturing the relative topographic relief of the Earth's surface. Under satellite nadir imaging, the relative heights and spatial continuity between objects remain invariant to sensor changes, making rDepth a domain-invariant feature for cross-domain semantic segmentation. Though pioneering efforts have incorporated rDepth for 3-D reconstruction [33], semantic segmentation [34], [35], change detection [36], DEM super-resolution [37], and hyperspectral classification [38], its application to cross-domain semantic segmentation of satellite remote sensing images remains underexplored. While certain RGB-T [39], [40], [41] and RGB-D [42], [43], [44] approaches have explored multimodal fusion of optical and depth information, they lack architectures tailored for cross-domain semantic segmentation in satellite imagery.

This article proposes a unified framework that leverages rDepth from DAM as an object-space geometric prior for cross-domain semantic segmentation. The framework is inherently extensible: its encoders, decoder, and fusion modules can be replaced with stronger alternatives. To isolate the contribution of rDepth, we deliberately use simple concatenation for feature fusion. The accompanying modules are tailored to rDepth as a derivative modality, incorporating asymmetric cross-modal alignment and geometry-aware refinement inspired by multiview stereo reconstruction [45], [46].

In summary, the main contributions of this article are as follows.

1) A unified depth-guided framework is proposed for optical remote sensing cross-domain segmentation, which for the first time introduces object-space rDepth as a domain-invariant geometric prior into the segmentation pipeline. We employ a smaller-scale encoder for rDepth, reducing parameters compared with the optical branch while maintaining sufficient representational capacity for geometric feature extraction. The extracted rDepth features are then adaptively fused with optical image features and fed into the decoder for segmentation inference.
2) A multiscale contrastive learning module is proposed to align optical and relative depth (rDepth) features. Specifically, an asymmetric weighted fusion strategy is introduced before contrastive optimization, preventing the rDepth-derived "half-modality" from dominating the learning process while enhancing optical feature representation.
3) A refinement strategy leveraging rDepth features is developed to guide semantic mask prediction. Specifically, inspired by geometric consistency principles in multiview stereo reconstruction, a compact fully convolutional module is utilized to encode the shallow features of rDepth. These encoded features are then incorporated as residuals of semantic probabilities via learnable parameters to optimize the initial semantic logits.

Unlike unsupervised domain adaptation (UDA), this work operates under the DG paradigm, which requires no target-domain images during training. This is a stricter and more practically applicable setting than UDA, yet our framework matches or exceeds the accuracy of UDA methods that rely on target-domain data. The remainder of this article is organized as follows. Section II details the proposed framework and modules. Section III presents the experimental results. Section IV analyzes model complexity and generalization to more challenging scenarios. Section V concludes the article.

II. Methodology

This section details the proposed depth-guided cross-domain semantic segmentation (DG-CroSeg) framework for satellite optical remote sensing images. As illustrated in Fig. 2, the framework features a dual-encoder design, multiscale contrastive alignment, and a depth-aware refinement module, taking optical remote sensing images and their zero-shot relative depth maps as inputs to enhance cross-DG.

A. Overall Framework

Let $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$ denote an input optical remote sensing image and $\mathbf{D} \in \mathbb{R}^{H\times W}$ denote the corresponding zero-shot rDepth map generated by DAM [30]. The core design of DG-CroSeg comprises three components, as illustrated in Fig. 2.

1) *Dual-Encoder Feature Extraction:* The optical image $\mathbf{I}$ is fed into the main encoder to extract optical features $\{\mathbf{F}_I^1, \mathbf{F}_I^2, \mathbf{F}_I^3, \mathbf{F}_I^4\}$ at four scales. Meanwhile, the rDepth

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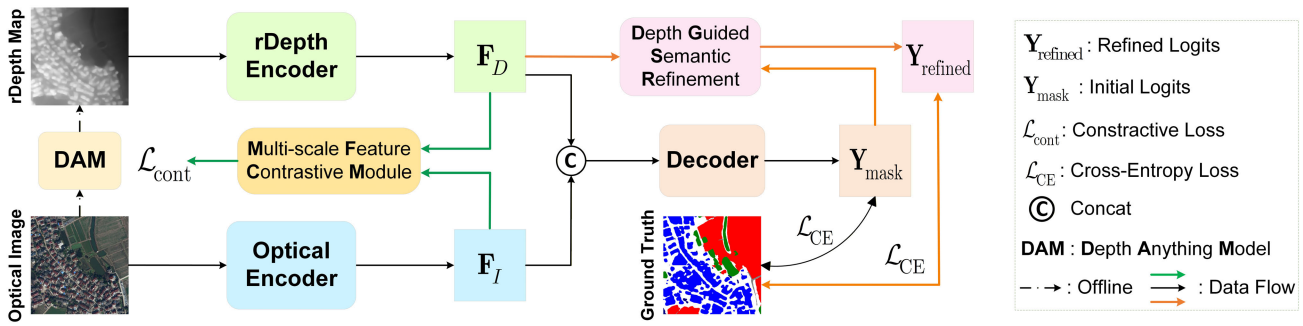


Fig. 2. Overview of the proposed relative DG-CroSeg for high-resolution satellite remote sensing images. Details of MFCM and DGSR described in Sections II-C and II-D.

map $\mathbf{D}$ is fed into a smaller dedicated encoder to extract rDepth features $\{\mathbf{F}_D^1, \mathbf{F}_D^2, \mathbf{F}_D^3, \mathbf{F}_D^4\}$ at four scales, where these rDepth features match the spatial resolutions of the optical features.

2) *Multiscale Feature Contrastive Module:* The multiscale contrastive loss is computed using the four-scale optical and rDepth feature pairs to enhance the modal consistency. Additionally, features at the same scale (e.g., $\mathbf{F}_I^1$ and $\mathbf{F}_D^1$) are concatenated along the channel dimension to form fused multimodal features.
3) *Depth-Guided Semantic Refinement:* The fused features are fed into the multiscale decoder (such as the UperNet [47]) for multiscale decoding, generating an initial semantic probability map. Subsequently, the depth-aware semantic optimization module leverages features extracted from rDepth to further refine the initial probability map.

The goal of DG-CroSeg is to learn a mapping function $F : (\mathbf{I}, \mathbf{D}) \mapsto \mathbf{S}_{\text{final}}$, where $\mathbf{S}_{\text{final}} \in \mathbb{R}^{H\times W\times C}$ is the refined semantic segmentation mask with C classes.

The proposed framework is fully compatible with most semantic segmentation models following the encoder–decoder paradigm (Fig. 2), serving as a unified rDepth-guided framework for cross-domain semantic segmentation of satellite optical remote sensing imagery.

B. Dual-Encoder Feature Extraction

Since DAM is pretrained on ground-level natural images, it tends to generate locally coarse relative depth estimations when directly applied to satellite nadir remote sensing imagery. Nevertheless, such local estimation deviations barely compromise the effectiveness of the rDepth prior, because our framework exclusively exploits the ordinal spatial structure of rDepth rather than its absolute metric precision. Specifically, DAM generates relative depth maps normalized to $[0, 1]$, which only encode surface height ranking instead of genuine physical elevation. Even when absolute elevation predictions for ground objects deviate, the inherent relative height order across land covers remains consistent. For instance, building rooftops are higher than surrounding terrain, water surfaces are lower than adjacent banks, and roads maintain a consistently flat relative topography. Thus, the rDepth could be an intuitive spatial prior for cross-domain semantic segmentation.

To balance representation capability and computational efficiency, we adopt two distinct encoders: a larger, high-capacity encoder for optical images and a compact encoder for rDepth maps.

Given an optical image $\mathbf{I}$, it is normalized and then fed into Encoder_{Opt} to output four-scale feature maps: $\{\mathbf{F}_I^1, \mathbf{F}_I^2, \mathbf{F}_I^3, \mathbf{F}_I^4\} = \text{Encoder}_{Opt}(\mathbf{I})$, where $\mathbf{F}_I^k \in \mathbb{R}^{H_k\times W_k\times C_I^k} (k = 1, 2, 3, 4)$, and the spatial resolution satisfies $H_k = H/2^k$.

Given a single-channel rDepth map $\mathbf{D}$ with fewer textures, a smaller encoder extracts features while maintaining computational efficiency. To match the three-channel input format required by pretrained models, $\mathbf{D}$ is expanded to three channels by repeating along the channel dimension. It is then fed into the corresponding encoder to obtain $\{\mathbf{F}_D^1, \mathbf{F}_D^2, \mathbf{F}_D^3, \mathbf{F}_D^4\} = \text{Encoder}_D(\mathbf{D})$, where $\mathbf{F}_D^k \in \mathbb{R}^{H_k\times W_k\times C_D^k} (k = 1, 2, 3, 4)$, and H_k, W_k are consistent with those of $\mathbf{F}_I^k$ to ensure scale compatibility.

C. Multiscale Feature Contrastive Module

The core motivation for multiscale feature contrastive module (MFCM) stems from a fundamental property of rDepth. As a derived representation rather than an independent modality, rDepth is computed entirely from the optical image. Conventional multimodal contrastive learning assumes two complementary information sources and treats them symmetrically. However, such an assumption fails for rDepth. If the derivative modality is allowed to dominate, optical features would regress toward depth-derived representations, thus losing rich spectral and textural cues essential for semantic discrimination. To address this issue, MFCM introduces an asymmetric fusion weight that explicitly subordinates rDepth to optical features. In this way, the depth prior only serves to reinforce rather than overshadow the optical representation.

As illustrated in Fig. 3, the multiscale contrastive module enhances the consistency between optical and rDepth features across four scales. Unlike conventional contrastive learning that treats modalities equally, this module introduces a learnable fusion weight to adaptively integrate features before contrastive optimization, while adopting memory-efficient designs.

The module operates on four-scale feature pairs $(\mathbf{F}_I^k, \mathbf{F}_D^k)(k = 1, 2, 3, 4)$, where $\mathbf{F}_I^k \in \mathbb{R}^{B\times C_I^k\times H_k\times W_k}$ and $\mathbf{F}_D^k \in \mathbb{R}^{B\times C_D^k\times H_k\times W_k}$. The core principle is to maximize the similarity of positive

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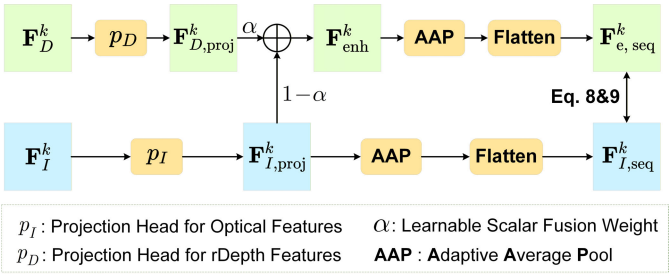


Fig. 3. Demonstration of MFCM.

pairs from the same sample and spatial position, while minimizing the similarity of negative pairs from different spatial positions.

Due to the inconsistent channel dimensions of optical and rDepth features ($C_I^k \neq C_D^k$), two independent projection heads are designed to map the features to a unified embedding space $\mathbb{R}^{B\times C_{\text{proj}}\times H_k'\times W_k'}$

$$\mathbf{F}_{I,\text{proj}}^k = p_I\left(\mathbf{F}_I^k\right), \quad \mathbf{F}_{D,\text{proj}}^k = p_D\left(\mathbf{F}_D^k\right) \tag{1}$$

where p_I and p_D denote the projection heads for optical and rDepth features, respectively. Each projection head consists of a 1×1 convolution to adjust channel dimension, batch normalization, and optional ReLU activation.

To adaptively balance the contribution of optical and rDepth features (critical for remote sensing scenes with varying depth quality), a learnable scalar fusion weight $\alpha \in [0,1]$ is introduced. The weight is parameterized as a logit α_{logit} and mapped to $[0,1]$ using the sigmoid function $\alpha = \sigma(\alpha_{\text{logit}})$, where $\alpha_{\text{logit}} \in \mathbb{R}$. Let the parameter α denote the weight of rDepth features and $1-\alpha$ denote the weight of optical features. The projected optical and rDepth features are then fused via element-wise addition with adaptive weights

$$\mathbf{F}_{\text{enh}}^k = (1-\alpha)\cdot\mathbf{F}_{I,\text{proj}}^k + \alpha\cdot\mathbf{F}_{D,\text{proj}}^k \tag{2}$$

where $\mathbf{F}_{\text{enh}}^k \in \mathbb{R}^{B\times C_{\text{proj}}\times H_k'\times W_k'}$ is the enhanced feature at scale k. This strategy integrates modality information before contrastive learning, enabling the model to learn unified cross-modal representations.

Since rDepth serves as prior guidance, optical features should play a dominant role. We therefore formulate the contrastive computation between the enhanced features $\mathbf{F}_{\text{enh}}^k$ and the projected optical features $\mathbf{F}_{I,\text{proj}}^k$, rather than the rDepth features $\mathbf{F}_{D,\text{proj}}^k$. For each scale k, $\mathbf{F}_{\text{enh}}^k$ and $\mathbf{F}_{I,\text{proj}}^k$ are downsampled via adaptive average pooling and flattened into 3-D sequence tensors for contrastive pair construction

$$\mathbf{F}_{\text{e, seq}}^k = \text{Flatten}\left(\mathbf{F}_{\text{enh}}^k\right) \in \mathbb{R}^{B\times L\times C_{\text{proj}}} \tag{3}$$

and

$$\mathbf{F}_{I,\text{ seq}}^k = \text{Flatten}\left(\mathbf{F}_{I,\text{proj}}^k\right) \in \mathbb{R}^{B\times L\times C_{\text{proj}}} \tag{4}$$

where $L = (H_k/r)\times(W_k/r)$ denotes the length of the spatial sequence, where the parameter r is the downsampling ratio, corresponding to the total number of spatial positions per sample after down-sampling. Subsequently, contrastive pairs are constructed based on spatial positions to reinforce cross-modal consistency and are listed below.

1) *Positive Pairs:* For each spatial position $l(l = 1,\ldots,L)$ in the bth sample $(b = 1,\ldots,B)$, the positive pair is $(\mathbf{F}_{\text{e, seq}}^k(b,l,:), \mathbf{F}_{I,\text{ seq}}^k(b,l,:))$ derived from the enhanced and projected optical features at the same sample and spatial position. This encourages the enhanced feature to retain semantic consistency with the original optical feature while incorporating relative depth information.
2) *Negative Pairs:* For each positive pair, negative pairs are constructed by sampling M spatial positions from other locations in the same sample, yielding $(\mathbf{F}_{\text{e, seq}}^k(b,l,:), \mathbf{F}_{I,\text{ seq}}^k(b,l',:))$ where $l' \neq l$. This intrasample negative sampling avoids intersample interference and focuses on distinguishing spatial differences.

Feature normalization is applied along the channel dimension (C_{proj}) for each spatial position to ensure consistency in similarity calculation

$$\mathbf{F}_{\text{e, norm}}^k = \frac{\mathbf{F}_{\text{e, seq}}^k}{\left\|\mathbf{F}_{\text{e, seq}}^k\right\|_2}, \quad \mathbf{F}_{I,\text{norm}}^k = \frac{\mathbf{F}_{I,\text{seq}}^k}{\left\|\mathbf{F}_{I,\text{ seq}}^k\right\|_2}. \tag{5}$$

Then, the similarity of positive pairs is computed as the cosine similarity between the enhanced feature and the original projected optical feature at the same spatial position

$$s_{\text{pos}}^k(b,l) = \text{sim}\left(\mathbf{F}_{\text{e, norm}}^k(b,l,:), \mathbf{F}_{I,\text{norm}}^k(b,l,:)\right). \tag{6}$$

This measures the consistency between the enhanced and original optical features, ideally approaching 1.0.

The similarity of negative pairs is the cosine similarity between the enhanced feature at the target spatial position and the original projected optical feature at sampled negative spatial positions

$$s_{\text{neg}}^k(b,l,m) = \text{sim}\left(\mathbf{F}_{\text{enhanced, norm}}^k(b,l,:), \mathbf{F}_{I,\text{proj, norm}}^k(b,l_m,:)\right) \tag{7}$$

where $l_m(m = 1,\ldots,M)$ denotes the mth negative spatial position. To ensure training stability, all similarities are linearly mapped to $[0,1]$. Following [48] and [49], the contrastive loss at scale k is defined as:

$$\mathcal{L}^k = -\frac{1}{B\times L}\sum_{b=1}^{B}\sum_{l=1}^{L} \log\left(\frac{\exp\left(s_{\text{pos}}^k(b,l)/\tau\right)}{\exp\left(s_{\text{pos}}^k(b,l)/\tau\right)+\sum_{m=1}^{M}\exp\left(s_{\text{neg}}^k(b,l,m)/\tau\right)}\right) \tag{8}$$

where τ is the temperature parameter controlling the sharpness of the similarity distribution. The total multiscale contrastive loss is formulated as

$$\mathcal{L}_{\text{cont}} = \sum_{k=1}^{4} w_k\mathcal{L}^k \tag{9}$$

where w_k is the weight for each $\mathcal{L}^k$ and balances the contribution of contrastive loss and segmentation loss during training.

To reduce GPU memory consumption, three optimizations are implemented: adaptive average pooling to reduce spatial sequence length by r^2, batch-wise negative sampling with self-mask to avoid full $L \times L$ similarity matrix computation, and optional microbatch processing for large spatial sequences.

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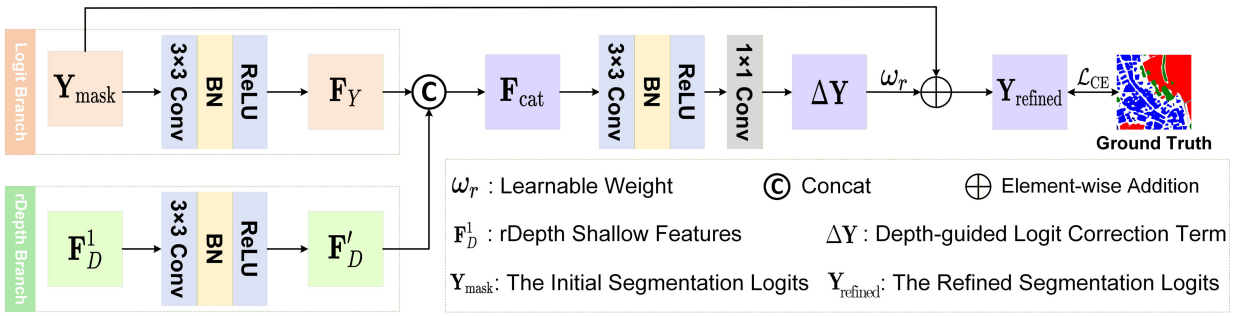


Fig. 4. Demonstration of DGSR module.

These strategies collectively reduce memory usage while preserving the contrastive learning signal.

The MFCM operates exclusively during training and incurs no computational cost during inference.

D. Depth-Guided Semantic Refinement

Deep semantic features already encode class-level information in the initial logits produced by the decoder. Adding deep rDepth features at this stage would reintroduce semantic ambiguity. In contrast, shallow rDepth features encode local geometric primitives such as edges, corners, and surface discontinuities. These structures are typically domain-invariant across different scenes. By using only shallow depth features as residual corrections to the semantic probability map, depth-guided semantic refinement (DGSR) refines boundary localization without overriding the semantic decisions already made by the optical branch.

Semantic segmentation in complex remote sensing scenes faces challenges at ambiguous boundaries and low-contrast regions, where optical features lack discriminative power. To address this, we propose the DGSR module, which utilizes relative depth features to adaptively refine initial segmentation logits. Its nonintrusive design enables seamless integration into most multiscale feature decoders without altering the original segmentation pipeline.

As illustrated in Fig. 4, the module takes two key inputs to realize depth-guided refinement, namely the initial semantic segmentation logits $\mathbf{Y}_{\text{mask}} \in \mathbb{R}^{B\times C\times H\times W}$, where B denotes batch size, C is the number of segmentation classes, and H, W represent spatial resolution. $\mathbf{Y}_{\text{mask}}$ is generated by the decoder's segmentation head, along with the shallow features of the relative depth map $\mathbf{F}_D^1$ extracted from the rDepth encoder. Shallow features from rDepth primarily encode edge and corner information. We therefore utilize only the shallow features from the rDepth encoder (Fig. 2) to refine $\mathbf{Y}_{\text{mask}}$. Here, $D_c \in \mathbb{N}^+$ denotes the channel dimension of rDepth features, configured to balance representation capability and computational efficiency while maintaining consistency with the semantic branch. Bilinear interpolation aligns the spatial resolution of the shallow rDepth features, yielding $\mathbf{F}'_D$.

To facilitate effective integration of semantic and geometric information, two parallel convolution branches are designed for feature extraction and dimension unification. The logit branch processes the initial segmentation logits $\mathbf{Y}_{\text{mask}}$ through a 3×3 convolution layer to adjust the channel dimension from C to D_c, followed by batch normalization (BN) and ReLU activation, which extracts high-level semantic feature embeddings while preserving spatial resolution via zero-padding. The rDepth branch adopts an identical structure to process the aligned rDepth features ($\mathbf{F}'_D$). This symmetric design ensures that semantic and geometric features reside in a unified feature space for effective cross-modal fusion.

The core operation generates depth-guided correction terms. The semantic features ($\mathbf{F}_Y$) and geometric features ($\mathbf{F}'_D$) are concatenated along the channel dimension

$$\mathbf{F}_{\text{cat}} = \text{Concat}\left(\mathbf{F}_Y, \mathbf{F}'_D\right) \in \mathbb{R}^{B\times 2D_c\times H\times W}. \tag{10}$$

The fused features pass through a two-stage convolution block. The first stage applies a 3×3 convolution to reduce channels from $2D_c$ to D_c with BN and ReLU. The second stage uses a 1×1 convolution to produce the depth-guided logit correction term $\Delta\mathbf{Y} \in \mathbb{R}^{B\times C\times H\times W}$, which targets ambiguous regions in the initial segmentation. To prevent overcorrection, a trainable scalar ω is mapped to $[0, 1]$ via the sigmoid function as $\omega_r = \sigma(\omega)$.

The refined segmentation logits $\mathbf{Y}_{\text{refined}}$ are obtained by adding the weighted correction term to the initial logits

$$\mathbf{Y}_{\text{refined}} = \mathbf{Y}_{\text{mask}} + \omega_r \cdot \Delta\mathbf{Y}. \tag{11}$$

This residual design preserves the initial semantic information while enabling depth-guided correction of ambiguous regions. The joint loss function supervises both the initial and refined logits

$$\mathcal{L}_{\text{sem}} = \mathcal{L}_{\text{CE}}\left(\mathbf{Y}_{\text{mask}}, \mathbf{G}\right) + \mathcal{L}_{\text{CE}}\left(\mathbf{Y}_{\text{refined}}, \mathbf{G}\right) \tag{12}$$

where $\mathbf{G}$ denotes the ground truth label. This ensures consistent gradient flow while guiding the model to learn depth-guided refinement patterns.

The contrastive loss in (9) is scaled by a coefficient γ ($\gamma = 1$ by default) and combined with (12), yielding the total loss

$$\begin{aligned}\mathcal{L}_{\text{total}} &= \mathcal{L}_{\text{sem}} + \gamma\mathcal{L}_{\text{cont}} \\ &= \mathcal{L}_{\text{CE}}\left(\mathbf{Y}_{\text{mask}}, \mathbf{G}\right) + \mathcal{L}_{\text{CE}}\left(\mathbf{Y}_{\text{refined}}, \mathbf{G}\right) + \gamma\sum_{k=1}^{4} w_k\mathcal{L}^k.\end{aligned} \tag{13}$$

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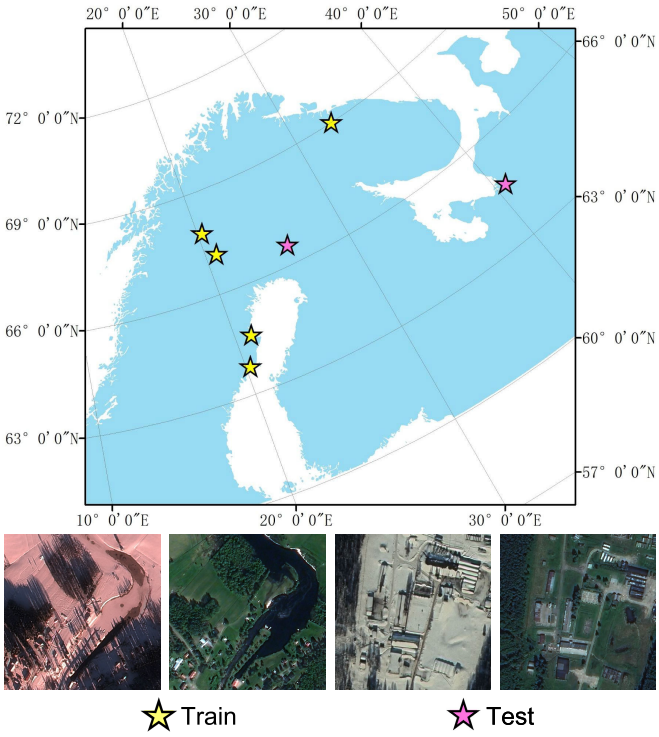


Fig. 5. Geographical location distribution of the SUSAN subset dataset, with example patches of the train and test set.

III. Experiments and Results

A. Datasets and Evaluation Metric

To evaluate our method on cross-domain semantic segmentation for high-resolution satellite remote sensing images, we use the **urban-to-rural** and **rural-to-urban** settings from the LoveDA dataset and **a self-constructed circumpolar cross-region and multiseason dataset**.

1) **LoveDA dataset** [50] is a benchmark for remote sensing image semantic segmentation and cross-domain adaptation tasks, encompassing urban and rural scenes across 18 representative administrative districts in Nanjing, Changzhou, and Wuhan. It consists of high-resolution spaceborne images with a 0.3 m ground resolution, covering a total area of over 536.15 km^2, with each image patch sized 1024 × 1024 pixels. The dataset provides 166 768 annotated semantic objects and 5987 image patches, labeled with seven classes (background, building, road, water, barren, forest, and agriculture). It is randomly partitioned into training, validation, and test sets with 2522, 1669, and 1796 patches, respectively, and additionally includes cross-domain subsets (**urban-to-rural** and **rural-to-urban**) to assess model generalization capability under scene distribution shifts. *All prediction results were submitted to the official platform for class-wise quantitative evaluation.*
2) **SUSAN dataset** (Spaceborne circUmpolar urban Submeter lAnd cover mappiNg) is further employed for validation, covering seven cities in Finland, Sweden, and Russia. Derived from Jilin-1 L3D satellite imagery with 0.75 or 0.5 m ground resolution, this dataset includes multiseasonal scenes with distinct characteristics: snow-covered scenarios, high contrast, and complex ground objects (see Fig. 5). It is annotated with nine classes, including three fine-grained classes (i.e., road subdivided into runway, taxiway and safety lane, and other road types) and five coarse-grained classes (building, water, bare land, vegetation, ice, and snow) with a background class. The dataset comprises 2484 image patches of 512 × 512 pixels, randomly split into training and test sets at a 3 : 2 ratio. No validation set is utilized during training, and inference is conducted using the final checkpoint of the trained model. To train some UDA models with the SUSAN dataset, we selected fewer than 100 image patches from the target domain without overlap with the test set, while these images remain inaccessible to all DG approaches.

As the LoveDA online evaluation server only provides class-wise intersection over union (IoU) and its mean value (mIoU), we report these two metrics for all experiments to ensure the consistency of experimental results. The IoU of the ith class is computed as

$$\mathrm{IoU}_i = \frac{TP_i}{(TP_i + FP_i + FN_i)} \tag{14}$$

where TP_i, TN_i, FP_i, and FN_i represent the number of true-positive, true-negative, false-positive, and false-negative samples of the ith class, respectively, and are derived from the confusion matrix. The mean intersection over union is then computed as

$$\mathrm{mIoU} = \frac{1}{C} \times \sum_{i=1}^{C} \mathrm{IoU}_i \tag{15}$$

where C is the number of segmentation classes.

B. Implementation Settings

The proposed method is implemented in PyTorch on a Linux system with an NVIDIA A100 GPU (40 GB memory). We compare with UDA methods including CLAN [51], AdaptSegNet [13], IAST [52], DAFormer [15], HRDA [24], MIC [16], ST-DASegNet [19], and S^{3}T-SFDA [21]. Several methods [17], [18], [20] are excluded as they only report results on self-partitioned test sets. For DG baselines, we evaluate popular encoder–decoder models with Swin [53], SegFormer [54], and ConvNeXt [55] as backbones paired with UperNet [47] and Mask2Former [56] as decoders. We adopt ConvNeXt-B as the optical backbone, whose parameter scale matches that of MiT-B5 employed in the recent UDA baselines above. A lighter ConvNeXt-S is used for rDepth encoding. All rDepth maps are generated by DAM V1 with a ViT-Large encoder.

Training is conducted with a batch size of 10 for 40 000 iterations. We use the AdamW optimizer with a learning rate of 0.0001, $(\beta_1, \beta_2) = (0.9, 0.999)$, and weight decay of 0.05. A two-stage learning rate schedule is adopted: linear warm-up from 10^{-6} during iterations 0–1500, followed by polynomial decay (power 1.0) to 0.0 over iterations 1501–40 000. Automatic mixed-precision (AMP) training with dynamic loss scaling is employed. Other configurations follow the defaults of [57].

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TABLE I

CROSS-DOMAIN RS IMAGE SEMANTIC SEGMENTATION COMPARISON RESULTS (%) FROM URBAN TO RURAL OF LOVEDA TEST DATASET. "AT" INDICATES ADVERSARIAL TRAINING, "ST" INDICATES SELF-TRAINING AND "DG" INDICATES DG. CONVNEXT ENCODERS WITH DIFFERENT SIZES (CONVN.: CONVNEXT, S: SMALL, B: BASE, XL: X-LARGE) ARE USED IN THIS EXPERIMENT. THE BEST PERFORMANCE IN EACH COLUMN IS HIGHLIGHTED IN BOLD, AND THE SECOND-BEST PERFORMANCE IN EACH COLUMN IS HIGHLIGHTED UNDERLINED

Method	Type	Modal	Class-wise IoU							mIoU
			Background	Building	Road	Water	Barren	Forest	Agriculture	
CLAN [51]	AT	Optical	22.93	44.78	25.99	46.81	10.54	37.21	24.45	30.39
AdaptSegNet [13]	AT	Optical	26.89	40.53	30.65	50.09	16.97	32.51	28.25	32.27
IAST [52]	ST	Optical	29.97	49.48	28.29	64.49	2.13	33.36	61.37	38.44
SegFormer [55]	DG	Optical	26.60	55.80	35.62	65.44	17.13	40.13	39.65	40.05
DCA [65]	ST	Optical	36.38	55.89	40.46	62.03	22.01	38.92	60.52	45.17
DAFormer [15]	ST	Optical	37.39	52.84	41.99	72.05	11.46	46.79	61.27	46.25
HRDA [24]	ST	Optical	36.80	**64.14**	40.92	71.80	14.36	47.17	68.37	49.08
MIC [16]	ST	Optical	39.23	61.92	43.85	72.38	13.92	47.67	**68.87**	49.69
ST-DASegNet (SegFormer) [19]	AT+ST	Optical	36.78	59.83	43.77	**73.83**	19.38	**49.96**	67.01	50.08
S^3T-SFDA [21]	ST	Optical	37.92	60.59	45.68	72.76	11.64	46.10	67.24	48.85
ConvN.-B (Baseline) [56]	DG	Optical	35.51	56.14	40.88	70.05	13.99	48.42	64.91	47.13
ConvN.-B (Optical)+ConvN.-S (rDepth)	DG	Optical+rDepth	36.79	59.27	**46.97**	71.31	19.87	41.62	67.05	48.98
ConvN.-XL (Optical)+ConvN.-B (rDepth)	DG	Optical+rDepth	37.23	63.05	42.69	71.71	19.91	45.03	68.36	49.71
ConvN.-B (Optical)+ConvN.-S (rDepth)+DG-CroSeg (ours)	DG	Optical+rDepth	**39.86**	63.77	41.44	71.15	**23.74**	45.40	68.32	**50.53**

As most parameters in our model are trainable, we set the temperature parameter $\tau = 0.07$ as suggested by MoCo V2 [58] and CLIP [59]. Since the contrastive loss is often considered as a regularization term for performance improvement and its weight generally ranges in $[0.01, 0.5]$ [60], [61], [62], the weights $w_k (k = 1, 2, 3, 4)$ of $\mathcal{L}^k$ are set to $w_1 = 0.02$, $w_2 = 0.02$, $w_3 = 0.03$ and $w_4 = 0.03$, respectively, in our experiments, while the ablation study of these parameters is conducted in Section III-G. To efficiently utilize memory while ensuring sufficient feature availability for constructing the contrastive learning loss, we set $C_{\text{proj}} = 128$ in all experiments.

C. Comparative Experiments

1) Results on the Urban-to-Rural Dataset: We first compare the proposed method with state-of-the-art UDA approaches and mainstream encoder–decoder-based semantic segmentation methods in the urban-to-rural scenario. Table I presents the class-wise IoU and the mean IoU (mIoU) of the overall segmentation results. Early adversarial training methods, including CLAN [51] and AdaptSegNet [13], fail to achieve satisfactory performance. IAST [52], a self-training method focusing on pseudo-label generation without the teacher–student architecture, is surpassed by SegFormer [54] and ConvNeXt-Base + UperNet [55], which benefit from well-pretrained backbones.

DCA [63] learns shared domain-invariant representations through class feature alignment, achieving an mIoU of 45.17%. DAFormer [15], HRDA [24], and MIC [16] share similar teacher–student self-training architectures with distinct optimizations: HRDA preserves fine-grained details for high-resolution images, while MIC incorporates consistency loss inspired by masked autoencoders [64] to leverage spatial context. All three outperform SegFormer [54], with HRDA and MIC also surpassing ConvNeXt-Base + UperNet [55]. ST-DASegNet [19] combines adversarial training with self-training and introduces feature-level adversarial learning, achieving an mIoU of 50.08% with 170.53 M parameters. S^3T-SFDA [21] strengthens target-domain learning through multiple dedicated modules with 97.23 M parameters, significantly outperforming early UDA methods.

ConvNeXt-Base + UperNet serves as our baseline. Even without rDepth, it attains strong DG due to high-quality pretraining. Adding the rDepth branch improves accuracy further, especially for buildings, where rDepth maps capture both surface continuity and edge discontinuities. Agriculture also benefits substantially: despite minimal depth variation, the class exhibits consistent rDepth patterns across large areas. Barren shows the largest gain, improving by 9.75% IoU. Bare surfaces appear as large continuous regions with slowly varying depth. rDepth captures this spatial continuity more reliably than optical texture, which fluctuates with lighting. Backbone scaling experiments confirm that larger models improve accuracy monotonically. With MFCM and DGSR, our model reaches 50.53% mIoU, surpassing all comparative methods and boosting both building and agriculture scores.

Visualization comparisons in Fig. 6 demonstrate superior performance in regions with sparse buildings and extensive natural landscapes. Boundary precision for buildings in Fig. 6(a) and (e) and vegetation in Fig. 6(c), (d), and (f) benefits from the DGSR module's depth-guided refinement, which captures subtle boundary details. Road continuity is improved through multiscale contrastive learning, which enhances consistent feature representations across scales. Although UDA methods misclassify lake regions in Fig. 6(c), our approach leverages consistent rDepth patterns of water surfaces to achieve more accurate segmentation. For slender or small objects such as narrow roads in Fig. 6(f) and small lakes in Fig. 6(c), our method achieves more effective detection by exploiting relative depth changes in transition regions. Adjacent buildings in Fig. 6(f) are clearly distinguished through relative depth analysis of edge gradients.

2) Results on the Rural-to-Urban Dataset: Table II presents the results, with key findings consistent with the urban-to-rural scenario. DAFormer [15], HRDA [24], and MIC [16] outperform early UDA methods. ST-DASegNet [19] and S^3T-SFDA [21], tailored for remote sensing cross-domain segmentation,

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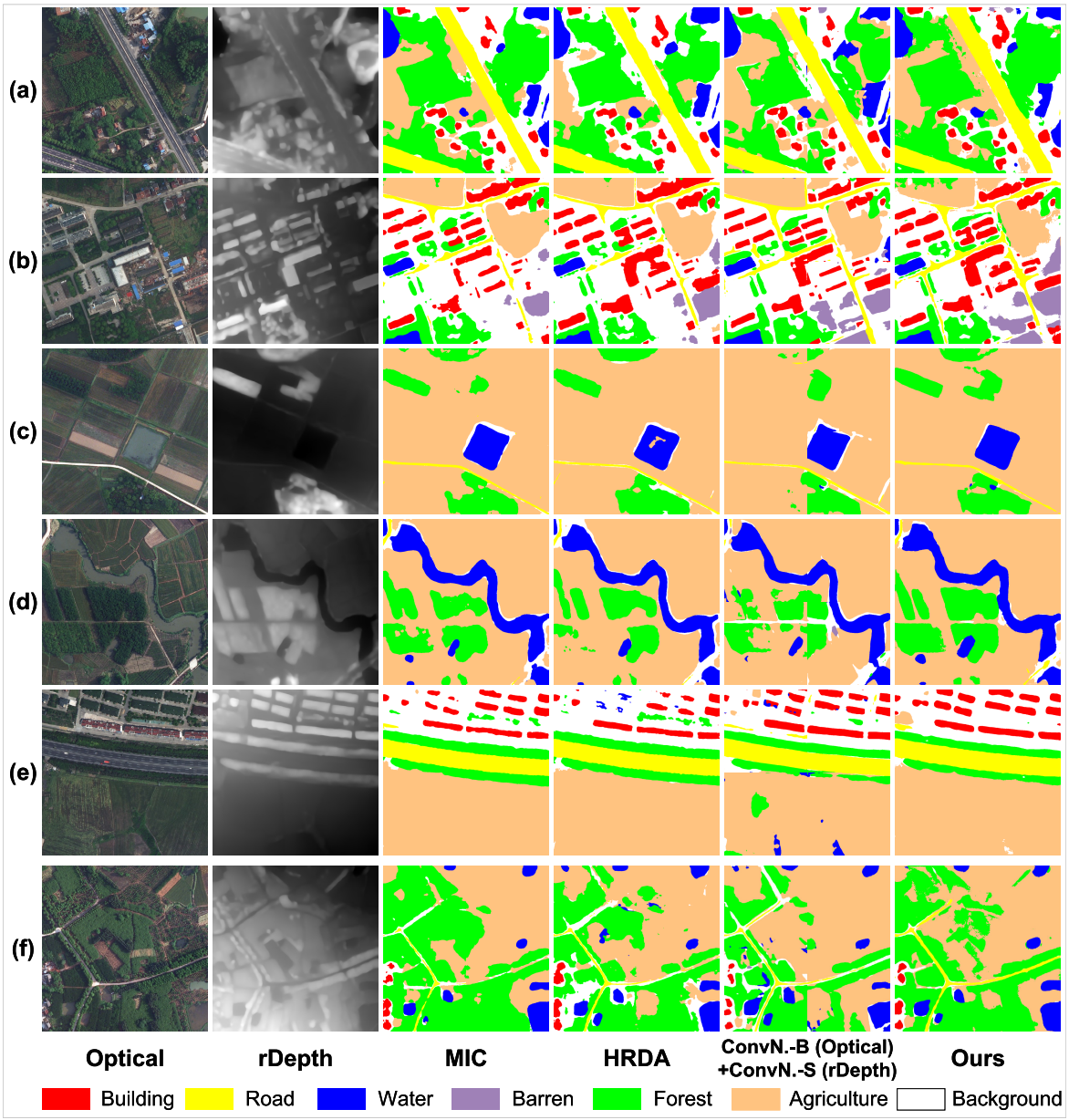


Fig. 6. Visualization comparisons from urban to rural on the LoveDA test set. (a)–(f) Denote different rural test scenes.

achieve superior performance. Integrating rDepth improves mIoU by 2.62%, and a larger backbone yields an additional 1% gain. With MFCM and DGSR, our method reaches 50.95% mIoU, comparable to S^3T-SFDA [21].

Notably, S^3T-SFDA [21] achieves an mIoU of 50.89%, significantly outperforming ST-DASegNet [19]. As noted in [21], this gap arises from class distribution biases between urban and rural scenes. In contrast, our method is not constrained by such biases, further validating its effectiveness.

The rural-to-urban transfer exposes models trained on rural scene to urban scene. rDepth aids this transition by encoding geometric properties that remain stable across the rural–urban divide. Road segmentation improves significantly, rising by 17.47% IoU. Rural roads generally exhibit a flat depth profile at near-ground elevation. The model therefore generalizes road geometry despite drastic changes in surface material and width. Building segmentation also rises substantially by 9.12% IoU. rDepth captures the vertical elevation discontinuity between rooftops and surrounding terrain. This structural signature is independent of roofing material or urban density. Agriculture registers a strong gain of 13.07% IoU. The subtle topographic microrelief produces fewer depth variations. These variations distinguish agricultural fields from flat impervious urban surfaces.

Visualization comparisons in Fig. 7 demonstrate strong transfer ability in regions with higher building density and complex urban infrastructure. Our method achieves sharper boundaries and accurately segments hollow areas within buildings in Fig. 7(f) by exploiting abrupt depth discontinuities at object edges and internal structural gaps. In transition regions such as Fig. 7(a), more precise class recognition is achieved through depth-guided sensitivity to gradual depth

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TABLE II

CROSS-DOMAIN RS IMAGE SEMANTIC SEGMENTATION COMPARISON RESULTS (%) FROM RURAL TO URBAN OF LOVEDA TEST DATASET. "AT " INDICATES ADVERSARIAL TRAINING, "ST" INDICATES SELF-TRAINING AND "DG" INDICATES DG. CONVNEXT ENCODERS WITH DIFFERENT SIZES (CONVN.: CONVNEXT, S: SMALL, B: BASE, XL: X-LARGE) ARE USED IN THIS EXPERIMENT. THE BEST PERFORMANCE IN EACH COLUMN IS HIGHLIGHTED IN BOLD, AND THE SECOND-BEST PERFORMANCE IN EACH COLUMN IS HIGHLIGHTED UNDERLINED

Method	Type	Modal	Class-wise IoU							mIoU
			Background	Building	Road	Water	Barren	Forest	Agriculture	
CLAN [51]	AT	Optical	43.41	25.42	13.75	79.25	13.71	30.44	25.80	33.11
AdaptSegNet [13]	AT	Optical	42.35	23.73	15.61	81.95	13.62	28.70	22.05	32.68
IAST [52]	ST	Optical	48.57	31.51	28.73	86.01	**20.29**	31.77	36.50	40.48
SegFormer [55]	DG	Optical	44.98	43.93	27.46	85.82	16.24	37.02	30.48	40.85
DCA [65]	ST	Optical	45.82	49.60	51.65	80.88	16.70	42.93	36.92	46.36
DAFormer [15]	ST	Optical	50.94	**56.66**	**62.83**	**89.41**	11.99	45.81	25.26	48.99
HRDA [24]	ST	Optical	48.25	45.24	59.16	87.17	18.82	44.94	27.61	47.31
MIC [16]	ST	Optical	50.66	49.56	49.46	87.87	19.51	43.48	34.66	49.31
ST-DASegNet (SegFormer) [19]	AT+ST	Optical	**51.01**	54.23	60.52	87.31	15.18	47.23	36.26	50.28
S^3T-SFDA [21]	ST	Optical	46.15	56.50	60.12	84.78	15.18	**48.29**	44.23	50.89
ConvN.-B (Baseline) [56]	DG	Optical	46.77	46.52	37.27	82.30	17.33	42.34	36.27	44.11
ConvN.-B (Optical)+ConvN.-S (rDepth)	DG	Optical+rDepth	48.23	52.68	42.70	77.18	16.62	40.85	48.86	46.73
ConvN.-XL (Optical)+ConvN.-B (rDepth)	DG	Optical+rDepth	50.56	51.81	42.89	85.11	16.69	42.67	44.00	47.68
ConvN.-B (Optical)+ConvN.-S (rDepth)+DG-CroSeg (ours)	DG	Optical+rDepth	**50.94**	55.64	54.74	84.98	17.47	44.05	**49.34**	**50.95**

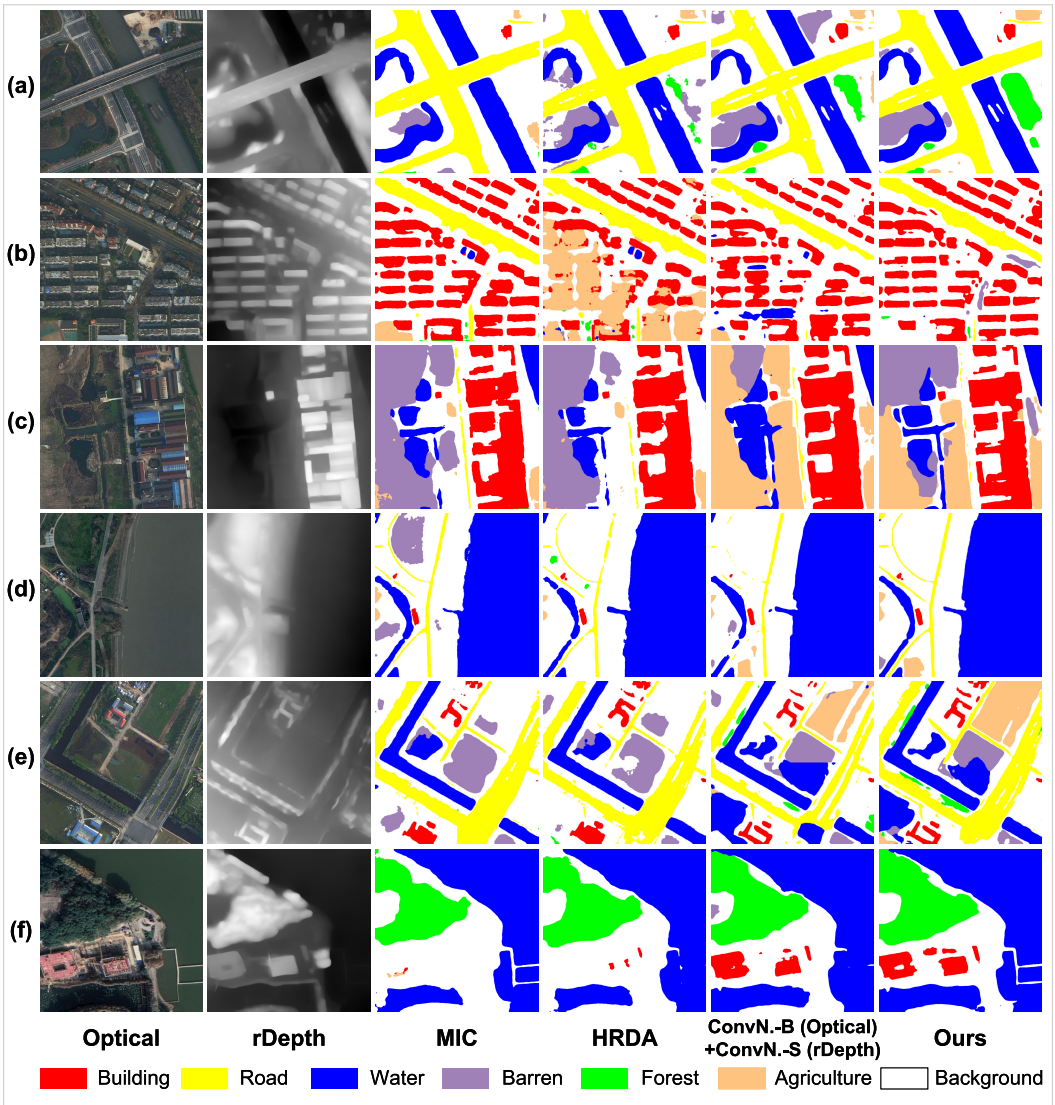


Fig. 7. Visualization comparisons from rural to urban on the LoveDA test set. (a)–(f) Denote different urban test scenes.

changes between adjacent land cover types. While ConvN.-B (Optical) + ConvN.-B (rDepth) misclassifies building shadows as water in Fig. 7(b), our method avoids this error by distinguishing depth variations: shadows exhibit minimal depth changes relative to buildings, whereas water bodies show distinct depth transitions. Previously undetected objects

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TABLE III

CROSS-DOMAIN RS IMAGE SEMANTIC SEGMENTATION COMPARISON RESULTS (%) ON SUSAN SUBSET TEST SET. "*" MEANS USING BOTH OPTICAL AND RDEPTH MODALS, METHODS WITHOUT "*" MEANS ONLY USING THE OPTICAL MODAL

Method	Type	Class-wise IoU									mIoU
		Fine-grained class (road)			Background	Coarse-grained classes					
		Runway	Taxiway and safety lane	Other road		Vegetation	Water	Bare land	Building	Ice and snow	
Swin-base [54]	DG	62.08	11.50	34.65	37.19	89.64	33.11	51.69	44.68	42.38	45.21
SegFormer (MiT-B5) [55]	DG	47.19	10.81	33.27	37.26	90.26	43.58	50.57	43.17	46.84	44.77
Mask2Former (ConvN.-B) [58]	DG	49.76	20.00	30.78	38.53	**90.94**	35.38	50.30	**49.72**	**49.81**	46.13
Mask2Former (Swin-base) [58]	DG	49.08	18.38	**37.06**	31.60	89.65	**50.82**	51.05	46.15	45.06	46.54
DAFormer [15]	ST	46.27	14.63	36.27	36.62	86.58	46.90	53.45	42.19	33.74	44.07
HRDA [24]	ST	55.36	15.90	33.47	32.91	90.22	44.58	52.82	39.15	47.57	45.59
MIC [16]	ST	53.86	21.25	31.85	37.44	88.45	48.96	50.96	40.82	37.73	45.70
ConvN.-B (baseline) [56]	DG	49.71	14.72	36.10	32.57	90.36	27.38	50.19	38.61	46.91	42.95
ConvN.-B (Optical)+ConvN.-S (rDepth)*	DG	67.56	17.47	35.19	34.02	89.09	29.72	51.01	48.24	42.07	46.04
ConvN.-XL (Optical)+ConvN.-B (rDepth)*	DG	66.35	14.45	35.83	37.21	89.65	39.06	52.28	46.44	42.91	47.13
ConvN.-B (Optical)+ConvN.-S (rDepth)+DG-CroSeg (ours)*	DG	**68.25**	**25.45**	37.00	**40.14**	89.26	32.54	**55.22**	47.24	43.77	**48.76**

such as vegetation in Fig. 7(a) and buildings in Fig. 7(f) are identified by leveraging unique depth signatures overlooked by the optical modality alone.

3) Results on the SUSAN Dataset: As shown in Table III, we benchmark against three open-source UDA methods: DAFormer [15], HRDA [24], and MIC [16]. Each surpasses the ConvNeXt-Base baseline, reaching 44.07%, 45.59%, and 45.70% mIoU. The extreme seasonal gap between winter snow cover and summer vegetation poses a severe challenge for self-training architectures, because pseudo-label quality drops when source and target appearances diverge sharply. In contrast, DAM infers relative topographic relief in a zero-shot manner for both seasons, providing a stable geometric reference independent of seasonal color shifts.

Consistent with LoveDA results, rDepth significantly enhances building segmentation accuracy with an IoU improvement of approximately 8% to 10%. The bare land class, analogous to agriculture in LoveDA, also benefits from consistent rDepth patterns across large areas. With MFCM and DGSR, mIoU reaches 48.76%, more than 3% higher than state-of-the-art UDA methods, demonstrating superior performance in complex cross-domain scenarios.

Geometrically, rDepth offers distinct advantages for classes where structural form dominates. Runway segmentation improves by approximately 18.5% IoU. Runways are elongated flat surfaces with uniform width. This regular geometric footprint remains stable in relative depth regardless of snow cover. The accuracy of taxiway and safety lane rises by over 10%. These connecting paths follow specific branching topologies. Their curvilinear transitions represent geometric traits that remain stable in rDepth even when optical texture changes. Bare land also benefits, improving by 5.03% IoU. Large bare areas exhibit consistent topographic signatures across seasons.

Visualization comparisons in Fig. 8 demonstrate strong transfer ability in intricate scenes around the Arctic Circle across seasons and regions. For fine-grained road subclasses (runway, taxiway and safety lane, and other road types) in Fig. 8(a), our method outperforms both DG and ST methods by leveraging rDepth to capture subtle geometric differences between road types such as width and surface continuity. In winter snow-covered scenes Fig. 8(c) and (d), our method identifies snow-covered buildings missed by other methods. Snow obscures RGB texture, but consistent rDepth signatures of buildings such as vertical edges and flat roofs remain distinguishable. Road continuity in cluttered environments Fig. 8(b) and (c) is maintained through MFCM, which aligns optical-rDepth-fused representations across scales, preserving global structural continuity while capturing local details. Sharper segmentation boundaries for buildings in Fig. 8(e) and small lakes in Fig. 8(b) are achieved through depth-guided refinement that highlights abrupt rDepth transitions at object boundaries.

D. Ablation Experiments of MFCM and DGSR

To analyze the contributions of the MFCM and the DGSR modules, we conducted ablation experiments across two benchmark datasets: the urban-to-rural (U2R) and the rural-to-urban (R2U). The results of these experiments, shown in Table IV, are presented in terms of class-wise IoU and mIoU.

Using MFCM alone yields mIoU of 49.59% on U2R and 48.99% on R2U, indicating that contrastive supervision improves feature learning but is insufficient for optimal segmentation.

Using DGSR alone achieves mIoU of 49.53% on U2R and 49.06% on R2U, confirming the value of refining initial segmentation masks with shallow rDepth features. DGSR alone improves building and road segmentation on both transfer directions. Shallow depth features precisely locate roof edges and road boundaries while optical features often blur these boundaries across domains. Barren shows a particularly strong gain on U2R, improving by 6.06% IoU. Shallow features capture gradual depth transitions at bare surface boundaries. Deep layers, by contrast, capture high-level semantic patterns. These patterns are already represented in the initial logits. Therefore, the deep layers contribute little complementary geometric information.

When both modules are combined, the model achieves its best performance: 50.53% on U2R and 50.95% on R2U. This demonstrates their complementary effects, as MFCM enhances optical-rDepth features before the decoder while DGSR refines predictions after the decoder.

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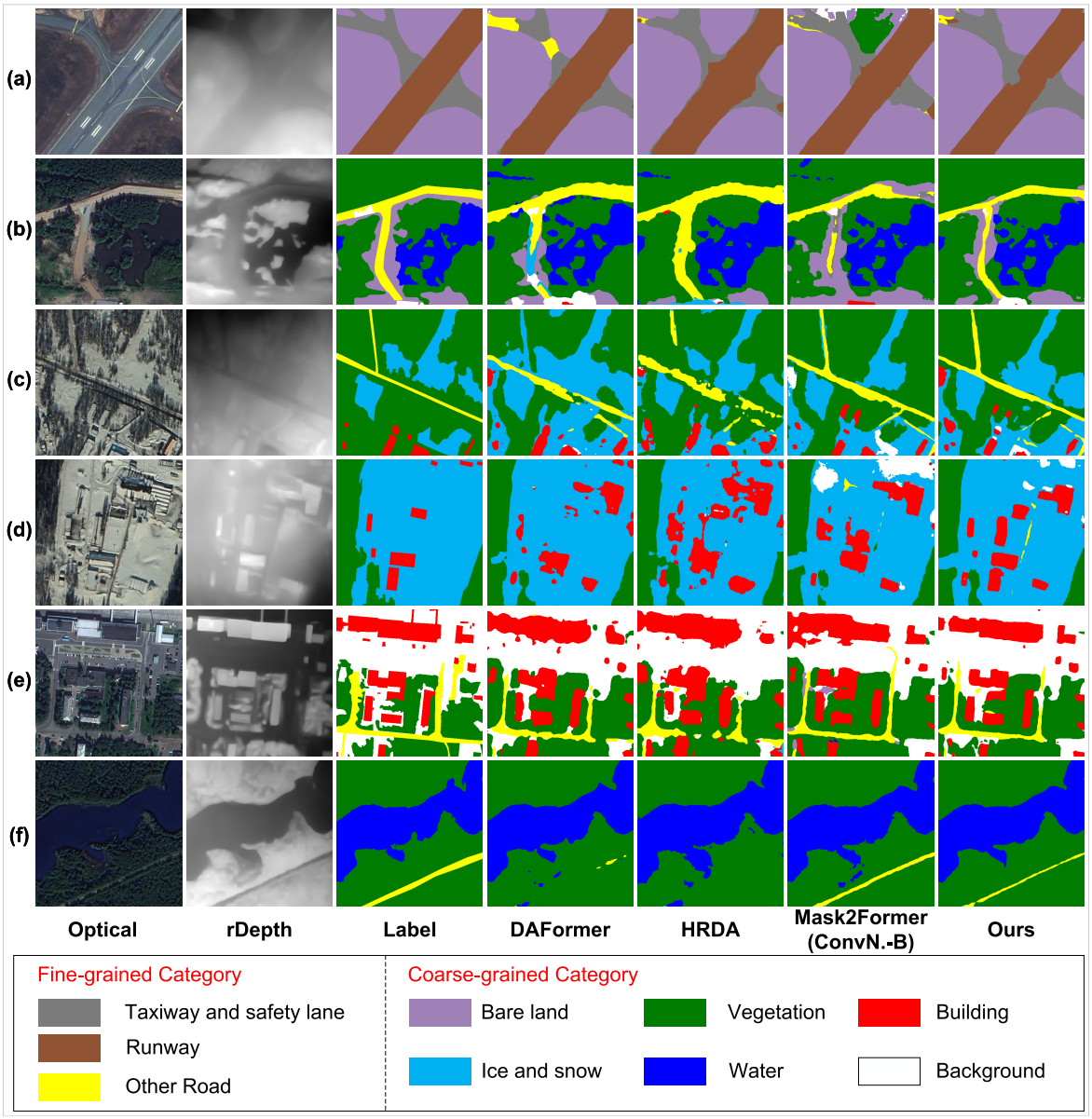


Fig. 8. Visualization comparisons on SUSAN subset test set. (a)–(f) Denote different cross-region and cross-season test scenes.

TABLE IV

RESULTS OF ABLATION EXPERIMENTS OF MFCM AND DGSR ON URBAN-TO-RURAL (U2R) AND RURAL-TO-URBAN (R2U) DATASETS. NUMERICAL COMPARISONS ARE PRESENTED IN FORM OF CLASS-WISE IOU (%)/MIOU (%), WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Method	Dataset	MFCM	DGSR	Class-wise IoU							mIoU
				Background	Building	Road	Water	Barren	Forest	Agriculture	
ConvN.-B (Optical)+ConvN.-S (rDepth)	U2R			35.51	56.14	40.88	70.05	13.99	**48.42**	64.91	47.13
ConvN.-B (Optical)+ConvN.-S (rDepth)	U2R	✓		35.23	**63.98**	40.57	**73.13**	21.12	45.76	67.30	49.59
ConvN.-B (Optical)+ConvN.-S (rDepth)	U2R		✓	36.93	60.03	**45.43**	72.45	20.05	45.26	66.56	49.53
ConvN.-B (Optical)+ConvN.-S (rDepth)	U2R	✓	✓	**39.86**	63.77	41.44	71.15	**23.74**	45.40	**68.32**	**50.53**
ConvN.-B (Optical)+ConvN.-S (rDepth)	R2U			48.23	52.68	42.70	77.18	16.62	40.85	48.86	46.73
ConvN.-B (Optical)+ConvN.-S (rDepth)	R2U	✓		50.30	53.52	45.41	**87.02**	**17.93**	**44.33**	44.45	48.99
ConvN.-B (Optical)+ConvN.-S (rDepth)	R2U		✓	49.09	53.67	48.44	84.01	17.42	41.60	49.16	49.06
ConvN.-B (Optical)+ConvN.-S (rDepth)	R2U	✓	✓	**50.94**	**55.64**	**54.74**	84.98	17.47	44.05	**49.34**	**50.95**

E. Ablation Experiments of Different Backbone Configurations

This ablation study investigates the impact of backbone networks with varying scales and architectures. We evaluate ConvNeXt [55] (Tiny, Small, Base, and xLarge), SwinTransformer [53] (Small, Base, and Large), and MiT series (MiT-B5, MiT-B1) from SegFormer [54], all with the UperNet decoder and our proposed MFCM and DGSR modules. Results on U2R are shown in Table V.

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TABLE V

RESULTS OF ABLATION EXPERIMENTS OF DIFFERENT BACKBONES ON URBAN-TO-RURAL (U2R) DATASETS. NUMERICAL COMPARISONS ARE PRESENTED IN THE FORM OF CLASS-WISE IOU (%)/MIOU (%), WHERE THE BOLD TEXT INDICATES THE BEST RESULTS. SWIN INDICATES THE SWINTRANSFORMER

No. of experiments	Backbone of optical images	Backbone of rDepth	Class-wise IoU							mIoU
			Background	Building	Road	Water	Barren	Forest	Agriculture	
1	ConvNeXt-Tiny	ConvNeXt-Tiny	36.57	59.07	46.90	70.01	16.49	45.04	66.63	48.67
2	ConvNeXt-Small	ConvNeXt-Tiny	38.43	63.18	45.06	70.74	19.25	44.81	67.34	49.83
3	ConvNeXt-Small	ConvNeXt-Small	38.53	63.67	42.57	70.98	20.99	45.29	**68.67**	50.10
4	ConvNeXt-Base	ConvNeXt-Small	**39.86**	63.77	41.44	71.15	**23.74**	45.40	68.32	50.53
5	ConvNeXt-XLarge	ConvNeXt-Base	38.50	**64.28**	43.91	**72.92**	20.63	45.46	68.32	**50.57**
6	MiT-B5	MiT-B1	33.40	60.99	42.41	70.59	19.79	**46.40**	65.83	48.49
7	Swin-Base	ConvNeXt-Small	35.28	55.90	**47.36**	70.34	19.22	44.70	65.78	48.37
8	Swin-Base	Swin-Small	34.85	57.40	45.14	70.71	18.31	44.11	66.53	48.15
9	Swin-Large	Swin-Small	36.43	58.08	47.32	72.75	20.45	44.80	68.05	49.70

TABLE VI

IMPACT OF LOSS WEIGHT SCALING FACTOR γ ON SEMANTIC SEGMENTATION MIOU (%) FOR CONVN.-B (OPTICAL) + CONVN.-S (RDEPTH) + DG-CROSEG MODEL ON U2R DATASET

	$\gamma = 1/2$	$\gamma = 1$	$\gamma = 2$	$\gamma = 3$	$\gamma = 4$	$\gamma = 5$
mIoU	50.27	50.53	50.92	50.66	50.46	50.18

The framework achieves the best performance with ConvNeXt as the backbone for both branches. CNNs inherently possess inductive bias toward local feature extraction, enabling their shallow layers to capture edge orientations and object shapes critical for DGSR refinement. In contrast, transformer-based architectures prioritize global contextual modeling, making their shallow layers less effective for fine-grained geometric features, as corroborated by the comparison between experiments No. 7 and No. 8. Additionally, CNNs exhibit faster convergence within a limited training budget.

Experiments No. 1 to No. 5 show that segmentation accuracy improves consistently with larger backbones. Even with small-scale ConvNeXt (No. 3), mIoU exceeds 50%, outperforming state-of-the-art UDA methods in Table I. These results confirm the effectiveness of incorporating rDepth guidance for cross-domain semantic segmentation.

F. Ablation Experiments of Contrastive Loss

The training involves two loss types in (13). To evaluate their proportional impact, we set $\gamma = 1/2, 1, 2, 3, 4, 5$ for training the *ConvN.-B (Optical) + ConvN.-S (rDepth) + DG-CroSeg* model on U2R. As shown in Table VI, all configurations achieve mIoU over 50.00%, demonstrating considerable robustness to lose weight variations.

G. Ablation Experiments of Different DAM Versions

To evaluate the impact of different DAM versions, we generate rDepth maps for the LoveDA dataset using DAM V1 [30], DAM V2 [31], and DAM V3 [32], all with ViT-Large encoders. Urban-to-rural cross-domain segmentation is then performed using ConvN.-B (Optical) + ConvN.-S (rDepth) + DG-CroSeg under identical settings. The generated rDepth maps are shown in Fig. 9, and the mIoU results are reported in Table VII.

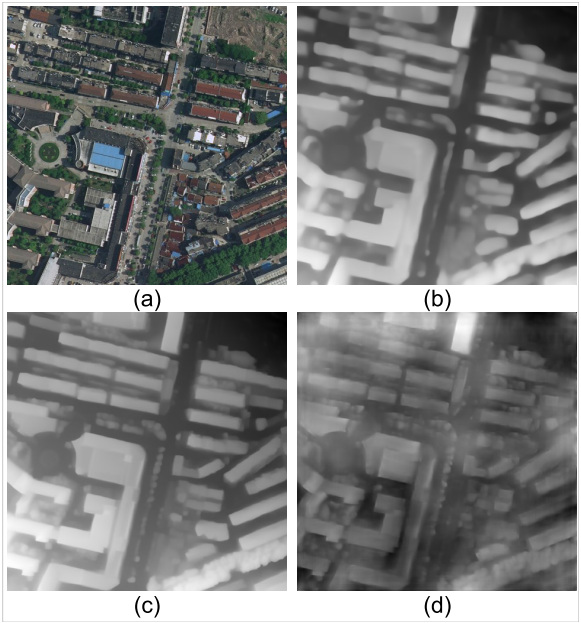


Fig. 9. Comparison of rDepth maps generated by different DAM versions. (a) Optical image. (b) DAM V1-large. (c) DAM V2-large. (d) DAM V3-large.

TABLE VII

IMPACT OF DIFFERENT DAM VERSIONS ON SEMANTIC SEGMENTATION MIOU (%) FOR CONVNEXT - BASE (OPTICAL) + CONVNEXT - SMALL (RDEPTH) + DG-CROSEG MODEL ON U2R DATASET

DAM Version	DAM V1	DAM V2	DAM V3
mIoU	50.53	50.56	49.86

All DAM versions effectively capture ground relief priors. DAM V3, a unified 3-D spatial foundation model for both relative depth generation and pose estimation, yields marginally lower segmentation accuracy than V1 and V2. Although V3 produces slightly less refined rDepth maps than the specialized V1 and V2 versions, it still provides adequate elevation priors for accurate cross-domain semantic segmentation.

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TABLE VIII

COMPARISON OF MODEL SIZE AND FLOPS FOR MODELS WITH DIFFERENT BACKBONES. THE SIZE OF THE INPUT IMAGE IS 3 × 512 × 512 PIXELS. BACKBONE_1 INDICATES THE BACKBONE OF OPTICAL IMAGE WHILE BACKBONE_2 INDICATES THE BACKBONE OF THE RDEPTH. SWIN INDICATES THE SWINTRANSFORMER

Backbone_1	FLOPs (GMacs) of Backbone_1	Backbone_2	FLOPs (GMacs) of Backbone_2	Total Parameters (M)	Total FLOPs (GMacs)	FPS
Data Preprocessing (Offline Depth Generation)						
DAM V1 (ViT-L)	414.83	–	–	332.68	589.99	32.02
Model Inference (Segmentation)						
ConvNeXt-Base	80.39	–	–	120.81	291.54	69.73
ConvNeXt-XLarge	318.61	–	–	388.67	532.89	46.97
ConvNeXt-Base	80.39	ConvNeXt-Small	45.49	176.29	341.50	45.67
ConvNeXt-XLarge	318.61	ConvNeXt-Base	80.39	484.40	618.88	33.46
Swin-Base	87.05	–	–	119.99	298.25	38.44
Swin-Large	194.23	–	–	231.88	406.93	37.78
Swin-Base	87.05	Swin-Small	49.38	174.85	352.12	22.18
Swin-Large	205.96	Swin-Small	49.39	287.06	472.58	22.07
MiT-B5	69.64	–	–	111.15	280.47	31.59
MiT-B5	69.64	MiT-B1	10.50	128.28	293.25	23.58

IV. DISCUSSION

A. Efficiency Analysis of DG-CroSeg

Table VIII compares model size, FLOPs, and inference speed (FPS) across various configurations with DG-CroSeg embedded in the UperNet decoder. All experiments are conducted under AMP with the input size fixed at 3 × 512 × 512. Incorporating rDepth increases parameters, but the FLOPs growth remains moderate: approximately 17.14% for ConvNeXt-Small, 16.14% for ConvNeXt-Base, 18.06% for Swin-Small, and only 4.56% for MiT-B1, as the MiT series is extensively optimized for efficiency. Regarding inference speed, the ConvNeXt-Base + ConvNeXt-Small configuration achieves 45.67 FPS, representing a moderate reduction from the optical-only baseline of 69.73 FPS. MiT-B5-based configurations yield lower FPS despite fewer parameters and FLOPs, as AMP retains more FP32 operations in self-attention to prevent numerical overflow, reducing the effective acceleration ratio compared to the purely convolutional ConvNeXt. Combined with Table V, the ConvNeXt series achieves the best tradeoff between accuracy, computational efficiency, and inference speed.

The dual-encoder design increases total parameters from 120.81 to 176.29 M relative to the optical-only baseline, and the rDepth encoder (ConvNeXt-S) is compact in comparison to the optical branch. This moderate overhead should be weighed against the accuracy gains reported in Tables I–III when considering deployment on resource-constrained platforms.

As shown in Table VIII, the DAM V1 (ViT-L) preprocessing achieves 32.02 FPS. Since DAM is executed only once per image as an offline preprocessing step and does not participate in forward or backward propagation, this cost is amortized and independent of training iterations. Table IX further presents computational complexity for different DAM versions and popular LVMs including SAM and DINO series, focusing on large-scale variants that are typically required for strong DG. Comparative analysis reveals that our ConvN.-B + ConvN.-S model comprises only 176.29 M parameters and 341.50 GMACs, lower than all large-scale ViT-based models while achieving competitive performance.

TABLE IX

COMPARISON OF DIFFERENT DAM VERSION COMPLEXITY. FLOPS ARE TESTED ON 3 × 512 × 512 WITH BATCH SIZE EQUALS 1 AS INPUT. VIT-H-L INDICATES HIERA LARGE VERSION OF VIT USED IN SAM V2. NOTE THAT SAM'S VIT-L USES PATCH SIZE 16 AND WINDOWED ATTENTION FOR EFFICIENCY, WHILE DINOV2'S VIT-L USES PATCH SIZE 14 AND GLOBAL SELF-ATTENTION OVER ALL TOKEN. SO, THE FLOPS OF BACKBONE MAY VARY BETWEEN DIFFERENT MODELS

Model	ViT type	FLOPs (GMacs) of Backbone	Total param.(M)	FLOPs (GMacs)
DAM V1 [30]	ViT-S	29.45	24.18	41.30
	ViT-B	117.08	96.10	159.51
	ViT-L	414.83	332.68	589.99
DAM V2 [31]	ViT-L	414.83	335.32	585.99
DAM V3 [32]	ViT-L	392.91	351.60	558.48
SAM V1 [67]	ViT-L	397.27	312.34	1495.66
SAM V2 [68]	ViT-H-L	202.62	224.45	813.06
DINO V2 [69]	ViT-L	475.56	338.65	688.72

B. DG From LoveDA Urban Dataset to WHU Building Dataset

To further validate generalizability, we conduct cross-dataset evaluation by applying the model trained on LoveDA Urban to the aerial subset of the WHU building dataset [68]. This subset presents a challenging scenario: it was captured with frame sensors rather than push-broom sensors, acquired in Christchurch, New Zealand, and features architectural styles that differ considerably from those in the Chinese urban areas covered by the LoveDA dataset.

LoveDA Urban labels are converted to binary classification (building versus background). We compare against three UDA approaches (DAFormer [15], HRDA [24], MIC [16]) and two RGB-T multimodal models (CAINet [39] and RoadFormer+ [40]). All methods are trained on LoveDA Urban and evaluated on the WHU test set. UDA methods use the WHU validation set as target domain data. All splits follow official releases.

The experimental results yield the following insights.

1) As indicated in Table X, our framework demonstrates superior performance even in this challenging cross-sensor scenario, benefiting from the effective fusion of optical and relative depth features. CAINet and

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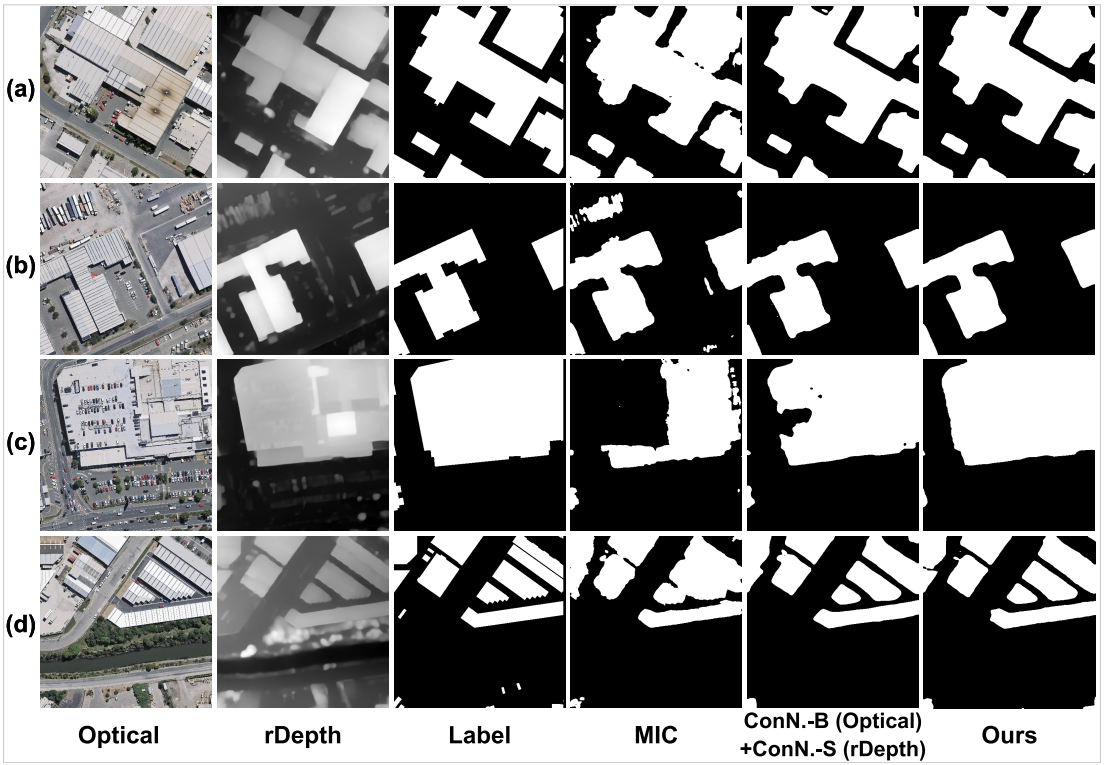


Fig. 10. Visualization comparisons from LoveDA Urban Dataset to WHU Building Dataset. (a)–(d) Denote different aerial building test scenes.

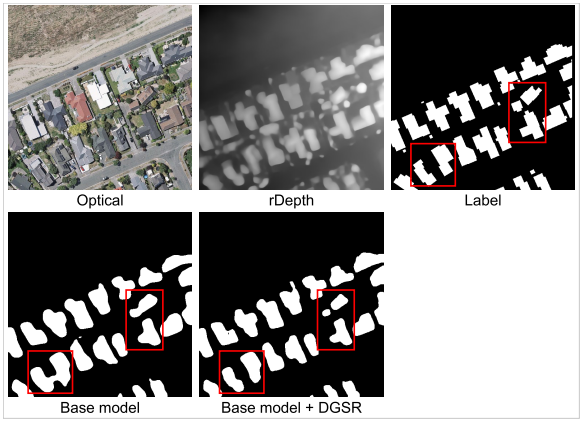


Fig. 11. Visualization comparisons of inference results w/ and w/o the DGSR module. The "base model" represents ConvN.-B (optical) + ConvN.-B (rDepth) without MFCM and DGSR.

RoadFormer+ are not designed for cross-domain tasks, resulting in inferior DG compared to UDA methods and our approach.

2) As shown in Fig. 10, relative depth effectively separates nearby buildings. Our method reduces misclassification errors and yields more refined boundaries, mitigating object adhesion. This further demonstrates the importance of DGSR and MFCM.
3) As evidenced by Figs. 11 and 12, shallow features contain geometric and directional information that enhances shape integrity and continuity in segmentation results.

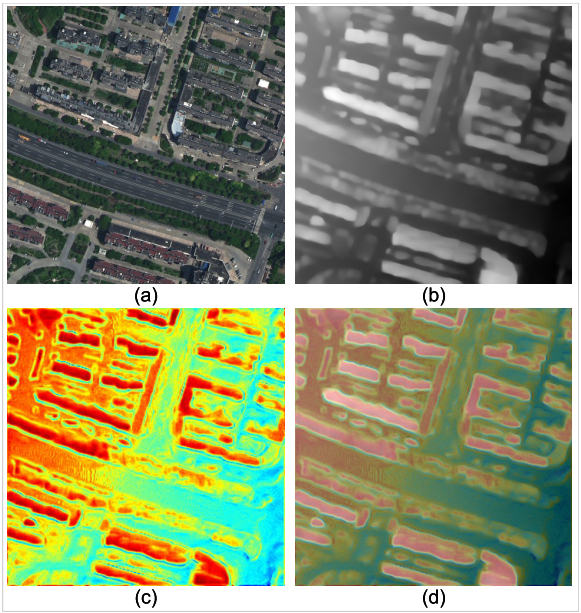


Fig. 12. Visualization of shallow relative depth features. (a) Optical image. (b) rDepth map generated by DAM. (c) Heatmap of the rDepth. (d) Heatmap overlaid on the original rDepth map.

C. Robustness Analysis Under Image Perturbations

We evaluate robustness by perturbing WHU test images during inference. Six settings are tested: Gaussian noise with $\sigma^2 \in \{0.01, 0.03, 0.06\}$, and combined Gaussian noise plus blur (3×3 kernel, $\sigma_b = 0.5$) at the same noise levels. All models are trained on clean LoveDA Urban data. For our

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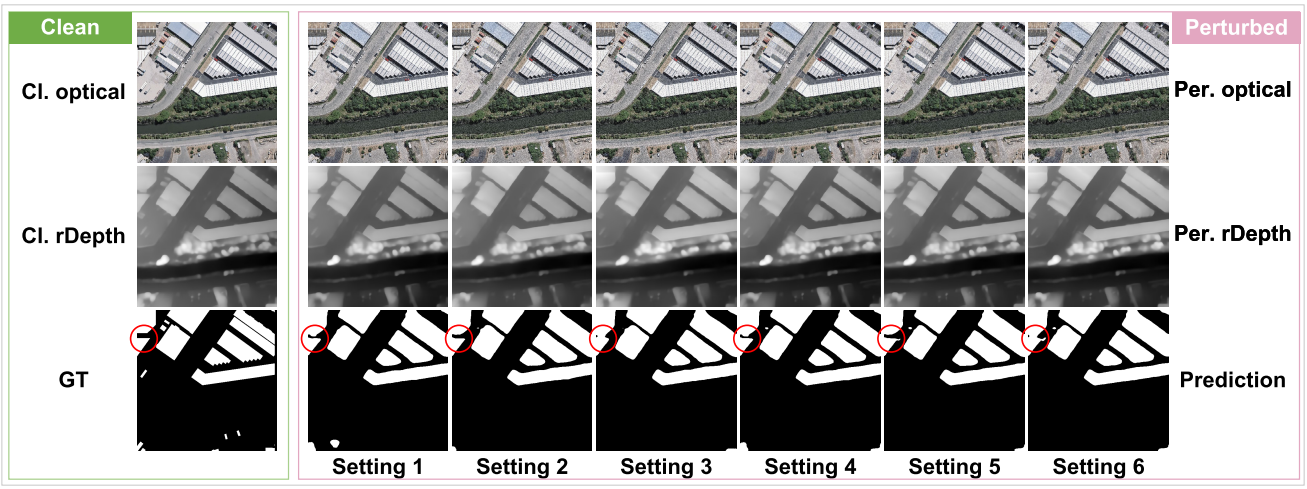


Fig. 13. Qualitative robustness analysis under image perturbations on the WHU Building dataset. Columns from left to right: clean image and six perturbation settings. Rows from top to bottom: clean/perturbed optical image, rDepth map from clean/perturbed optical image, ground truth/prediction under six settings. Cl. and Per. refer to clean and perturbed, respectively.

TABLE X

CROSS-DOMAIN RS IMAGE SEMANTIC SEGMENTATION COMPARISON RESULTS (%) FROM LOVEDA URBAN DATASET TO WHU BUILDING DATASET. "ST" INDICATES SELF-TRAINING AND "DG" INDICATES DG. THE BASE MODEL INDICATES CONVN.-B (OPTICAL) + CONVN.-S (RDEPTH) CONFIGURATION. THE BEST PERFORMANCE OF IOU IS HIGHLIGHTED IN BOLD

Method	Type	Modal	Building IoU
DAFormer [15]	ST	Optical	67.81
HRDA [24]	ST	Optical	73.05
MIC [16]	ST	Optical	74.05
DINOv2-Base [69]	DG	Optical	71.12
DINOv2-Large [69]	DG	Optical	72.31
SAM2.1-Base+ [68]	DG	Optical	73.26
SAM2.1-Large [68]	DG	Optical	74.58
CAINet [39]	DG	Optical+rDepth	67.06
RoadFormer+ [40]	DG	Optical+rDepth	71.66
Base model	DG	Optical+rDepth	74.29
Base model+DGSR	DG	Optical+rDepth	75.73
Base model+MFCM	DG	Optical+rDepth	75.83
Base model+DG-CroSeg (ours)	DG	Optical+rDepth	**76.41**

TABLE XI

ROBUSTNESS ANALYSIS: BUILDING IOU (%) UNDER DIFFERENT IMAGE PERTURBATIONS ON THE LOVEDA URBAN TO WHU BUILDING SETTING. GN DENOTES GAUSSIAN NOISE WITH VARIANCE σ^2, AND GB DENOTES GAUSSIAN BLUR (3×3 FOR KERNEL SIZE AND $\sigma_b = 0.5$). THE BEST RESULT IN EACH COLUMN IS HIGHLIGHTED IN BOLD

Method	Clean	GN only			GN + GB		
	–	Setting 1 σ^2=0.01	Setting 2 σ^2=0.03	Setting 3 σ^2=0.06	Setting 4 σ^2=0.01	Setting 5 σ^2=0.03	Setting 6 σ^2=0.06
DINOv2-Base	71.12	60.63	52.28	43.57	62.54	56.53	49.66
DINOv2-Large	72.31	63.46	57.31	49.72	63.40	57.72	50.93
SAM2.1-Base+	73.26	50.64	38.85	30.33	59.64	47.58	37.47
SAM2.1-Large	74.58	71.56	67.56	60.58	**72.86**	71.31	68.56
Ours	**76.41**	**72.80**	**70.57**	**67.54**	72.53	**71.31**	**68.84**

method, perturbed images are also fed into DAM to generate degraded rDepth maps, simulating a realistic scenario where input quality cannot be guaranteed.

Table XI reports building IoU for our method and four foundation-model baselines. Under mild noise ($\sigma^2 = 0.01$), our method retains 72.80%, outperforming DINOv2-Base by 12.17% and SAM2.1-Base+ by 22.16%. At $\sigma^2 = 0.06$, our method still reaches 67.54%, while DINOv2-Base falls to 43.57% and SAM2.1-Base+ to 30.33%. SAM2.1-Large shows the strongest baseline robustness, yet our method consistently leads. These results indicate that the rDepth geometric prior stabilizes predictions: even as texture and color cues degrade, the structural information in rDepth partially compensates. Under extreme noise, DAM-generated rDepth maps also degrade, reducing accuracy for all methods. The performance boundary is therefore tied to both input quality and DAM's zero-shot capability.

Fig. 13 visualizes the effect of different perturbations on our method. Across all settings, the rDepth maps maintain relatively stable overall structures, indicating the robustness of DAM's zero-shot inference. Under mild to moderate perturbations (Settings 1, 2, 4, 5), the predictions remain close to the ground truth. As perturbation intensity increases (Settings 3, 6), segmentation accuracy decreases due to the combined degradation of both optical features and rDepth quality. This confirms that the performance boundary of our framework is tied to the overall input quality rather than a single modality. Gaussian noise and blur serve as generic proxies, yet real-world satellite radiometric degradation (e.g., Poisson noise for photon-counting sensors, striping artifacts for push-broom scanners) is more complex. Future robustness studies could incorporate sensor-specific radiometric degradation models to better approximate operational conditions.

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V. Conclusion

This article proposes a unified depth-guided framework (DG-CroSeg) for cross-domain semantic segmentation of satellite remote sensing images. The compatible framework can flexibly integrate various encoders and decoders without architectural constraints. By employing relative depth as a domain-invariant spatial prior, our framework achieves strong generalization without relying on accessible target-domain optical data. This advantage enables practical deployment in satellite remote sensing data processing applications, including sustainable urban planning, dynamic city expansion monitoring, and fine-grained land cover mapping. Our proposed optical-rDepth paradigm reduces dependence on costly target-domain data collection and annotation, directly assisting satellite engineers in performing robust large-scale interpretation under inconsistent imaging conditions. Limitations include degraded performance in near-flat regions such as fields and deserts. This reflects an inherent constraint of the depth-guided prior. rDepth encodes relative surface height variation, and when topographic relief approaches zero, the resulting depth maps exhibit minimal contrast that conveys limited geometric information for semantic discrimination. In such cases, the model relies primarily on appearance cues, and the performance boundary is tied to the discriminative power of optical features rather than depth-guided priors. Insufficient handling of large cross-sensor resolution gaps is also noted.

Future work will focus on three directions: 1) optimizing DG-CroSeg as a plug-and-play component for fast inference; 2) designing resolution-adaptive alignment for multisatellite collaborative observation; and 3) exploring terrain-aware strategies that classify scene types into low-relief and high-relief regions at inference time, adjusting the model's reliance on optical versus depth cues accordingly.

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